RippleLogic Cascade Standard v1.0

Source-boundary rule: If this compact standard conflicts with the RippleLogic Canon, the Canon controls.

Release: MathGov Core Release 2026.09 v11.4 Complete Ready

This standard is the compact public reference for the current five-level RippleLogic cascade inside the MathGov framework. It prevents companion documents from repeating long synchronization notes.

TRC-CSV feedback rule. If CSV discovers a catastrophic, irreversible, or ruin-path scenario not represented in TRC, the run MUST reopen TRC before RLS. CSV does not absorb TRC.

Gate-critical confidence guard. Low confidence in adverse rights-covered, catastrophe-covered, or material CSV impacts cannot by itself make an option pass RF/NCRC, TRC, or CSV. If a gate outcome could change, the run must use a governed conservative bound, collect evidence and rerun, narrow the claim, downgrade, escalate, or refuse the stronger claim.

Baseline pointer. Gate-admissibility cells and residual welfare cells may use distinct baselines where the Canon requires it; see Canon §5.1A for the floor-reference versus status-quo dual-baseline rule.

Public cascade

RG -> RF -> TRC -> CSV -> RLS

Plain language: Ground reality -> protect rights -> bound ruin -> preserve the structure -> score the ripples.

Formal shorthand

RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS

- RG/RSG: Reality Grounding / Reality-Surface Grounding. Claim-authority precondition, not an ordinary option-rejecting gate.

RG status rule. Reality Grounding is Level 1 of the public method and a claim-authority precondition. It can force narrowing, escalation, exploratory-marking, or refusal, but it is not an option-rejecting gate; RF/NCRC, TRC, and CSV are the option-rejecting gates, and RLS ranks only what survives them. - RF/NCRC: Rights Floor, formally implemented by the Non-Compensatory Rights Constraint. Non-compensatory gate. - TRC: Tail-Risk Constraint. Non-compensatory catastrophic-risk gate. - CSV: Containment and Structural Viability. Selectability gate for containing-system integrity and execution viability. - RLS:

RippleLogic Score. Residual welfare-ranking layer applied only to options that remain selectable.

Computability vs realizability rule

A computed, simulated, generated, or model-fluent option is not automatically grounded, selectable, executable, or ethical. RG supplies claim authority; RF/NCRC supplies rights admissibility; TRC supplies ruin bounding; CSV supplies structural and execution viability; RLS ranks only the surviving selectable set.

Non-overlap rule

RG determines what claim may be made. RF/NCRC, TRC, and CSV determine whether an option remains selectable. RLS ranks only selectable options. RLS cannot rescue rights-floor failure, tail-risk failure, or CSV failure.

CSV non-purity rule

CSV does not reject every negative ripple. Negative ripples must be made visible and routed. Bounded residual harms may enter RLS. Uncontained, structurally degrading, unjustly externalized, non-viable, hidden, lock-in-producing, unmonitored, or beyond-tolerance harms require controls, redesign, escalation, emergency-provisional handling, or failure.

All-Encompassing Infinite Union (AIU) and SGP boundary

All-Encompassing Infinite Union (AIU) is a horizon/meta-union orientation, not a Tier 1-3 scoring object or override. SGP is a moral-status and protection evidence interface; it informs protected-stakeholder modeling where permitted but does not replace RG, RF/NCRC, TRC, CSV, RLS, lawful authority, or governance-role requirements.